Rounding and Significant Figures

Name _____ Date _____ No calculator Version 1

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KEY WORDS round *(whakaawhiwhi)* · decimal place *(mati ā-ira)* · significant figure *(mati hira)*

A Rounding to decimal places ●○○

Remember: Look at the next digit. If it is 5 or more, round up.

1. Round each number to the number of decimal places (d.p.) shown.

a) 3.476 (1 d.p.) = _____ **b)** 12.342 (2 d.p.) = _____

c) 0.8765 (3 d.p.) = _____ **d)** 7.95 (1 d.p.) = _____

e) 15.4849 (2 d.p.) = _____ **f)** 0.0637 (2 d.p.) = _____

g) 9.996 (2 d.p.) = _____ **h)** 24.5 (nearest whole number) = _____

2. Round 6.4807 to:

a) the nearest whole number = **b)** 1 d.p. = _____ **c)** 2 d.p. = _____

d) 3 d.p. = _____

B Counting significant figures ●○○

Remember: Zeros at the front of a number are never significant. Zeros between other digits always are.

3. How many significant figures does each number have?

a) 4,205 _____ **b)** 0.0031 _____ **c)** 7.50 _____ **d)** 0.0408 _____

e) 12.006 _____ **f)** 0.90 _____ **g)** 306 _____ **h)** 0.00070 _____

C Rounding to significant figures ●●○

Watch out: Keep the size of the number the same: 4,567 to 2 significant figures is 4,600, not 46.

4. Round each number to the number of significant figures (s.f.) shown.

a) 4,567 (2 s.f.) = ____ **b)** 0.03728 (2 s.f.) = ____ **c)** 28.46 (3 s.f.) = ____ **d)** 6,149 (1 s.f.) = ____

e) 0.9052 (3 s.f.) = ____ **f)** 153,790 (2 s.f.) = ____ **g)** 0.04996 (2 s.f.) = ____ **h)** 7.008 (2 s.f.) = ____

i) 362.47 (4 s.f.) = ____ **j)** 0.50382 (3 s.f.) = ____ **k)** 8,975 (3 s.f.) = ____ **l)** 0.007049 (2 s.f.) = ____

5. Round 45,678 to:

a) 1 s.f. = _____ **b)** 2 s.f. = _____ **c)** 3 s.f. = _____ **d)** 4 s.f. = _____

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D Estimating ●●○

Remember: Round each number to 1 significant figure, then calculate.

6. Estimate each answer by rounding the numbers to 1 significant figure first.

a) 38 × 52 **b)** 612 ÷ 29 **c)** 7.8 × 4.1 **d)** 4,870 + 3,215

e) 0.48 × 61 **f)** 893 ÷ 3.2 **g)** 19.7 × 5.2 **h)** 31 × 31

i) 203 × 4.9 **j)** 7.9 × 7.9 **k)** 5,890 ÷ 31 **l)** 0.91 × 68

E Problems ●●●

7. A crowd of 23,476 people watched a rugby match. Round this number to:

a) the nearest thousand **b)** 2 significant figures **c)** the nearest hundred

8. Tickets cost $38.50 each. Estimate the cost of 62 tickets by rounding each number to 1 significant figure.

9. A calculator shows 6.4807406. Write this number to:

a) 2 decimal places **b)** 3 significant figures **c)** 1 decimal place

10. Ari says that 0.0450 has 4 significant figures. Is he right? Explain your answer.

11. The population of a town is 18,649. Round this number to:

a) the nearest 10 **b)** the nearest 100 **c)** 1 significant figure

12. A car travels 317 km on 28.6 litres of petrol. Estimate how far it travels on each litre.

13. Sione rounds 4.96 to 1 decimal place and writes 5. What is wrong with his answer?

★ Extension

14. A length is 8 cm, rounded to the nearest centimetre. Write the smallest and largest values the length could be.

15. A mass is 4.6 kg, rounded to 1 decimal place. Write the smallest and largest values the mass could be.

16. A rectangle measures 6 cm by 4 cm, each rounded to the nearest centimetre. Find the smallest area it could have.

17. A number is 3,700 when it is rounded to 2 significant figures. What is the smallest the number could be?

18. How many whole numbers give 600 when they are rounded to the nearest hundred?

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Rounding and Significant Figures

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS round *(whakaawhiwhi)* · decimal place *(mati ā-ira)* · significant figure *(mati hira)*

A Rounding to decimal places ●○○

Remember: Look at the next digit. If it is 5 or more, round up.

1. Round each number to the number of decimal places (d.p.) shown.

a) 5.238 (1 d.p.) = _____ **b)** 8.467 (2 d.p.) = _____

c) 0.4321 (3 d.p.) = _____ **d)** 6.97 (1 d.p.) = _____

e) 23.7351 (2 d.p.) = _____ **f)** 0.0582 (2 d.p.) = _____

g) 4.998 (2 d.p.) = _____ **h)** 37.5 (nearest whole number) = _____

2. Round 3.7416 to:

a) the nearest whole number = **b)** 1 d.p. = _____ **c)** 2 d.p. = _____

d) 3 d.p. = _____

B Counting significant figures ●○○

Remember: Zeros at the front of a number are never significant. Zeros between other digits always are.

3. How many significant figures does each number have?

a) 6,037 _____ **b)** 0.0054 _____ **c)** 3.20 _____ **d)** 0.0709 _____

e) 25.003 _____ **f)** 0.60 _____ **g)** 807 _____ **h)** 0.00040 _____

C Rounding to significant figures ●●○

Watch out: Keep the size of the number the same: 4,567 to 2 significant figures is 4,600, not 46.

4. Round each number to the number of significant figures (s.f.) shown.

a) 8,342 (2 s.f.) = ____ **b)** 0.05284 (2 s.f.) = ____ **c)** 47.38 (3 s.f.) = ____ **d)** 2,951 (1 s.f.) = ____

e) 0.6038 (3 s.f.) = ____ **f)** 247,300 (2 s.f.) = ____ **g)** 0.03997 (2 s.f.) = ____ **h)** 9.004 (2 s.f.) = ____

i) 518.36 (4 s.f.) = ____ **j)** 0.70461 (3 s.f.) = ____ **k)** 6,485 (3 s.f.) = ____ **l)** 0.003052 (2 s.f.) = ____

5. Round 27,364 to:

a) 1 s.f. = _____ **b)** 2 s.f. = _____ **c)** 3 s.f. = _____ **d)** 4 s.f. = _____

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D Estimating ●●○

Remember: Round each number to 1 significant figure, then calculate.

6. Estimate each answer by rounding the numbers to 1 significant figure first.

a) 43 × 68 **b)** 795 ÷ 41 **c)** 6.2 × 8.9 **d)** 7,120 + 2,890

e) 0.52 × 78 **f)** 612 ÷ 1.9 **g)** 28.4 × 6.1 **h)** 48 × 48

i) 396 × 5.1 **j)** 6.8 × 6.8 **k)** 8,120 ÷ 39 **l)** 0.79 × 52

E Problems ●●●

7. A crowd of 47,832 people watched a rugby match. Round this number to:

a) the nearest thousand **b)** 2 significant figures **c)** the nearest hundred

8. Tickets cost $72.50 each. Estimate the cost of 38 tickets by rounding each number to 1 significant figure.

9. A calculator shows 3.7416573. Write this number to:

a) 2 decimal places **b)** 3 significant figures **c)** 1 decimal place

10. Mia says that 0.0306 has 4 significant figures. Is she right? Explain your answer.

11. The population of a town is 27,358. Round this number to:

a) the nearest 10 **b)** the nearest 100 **c)** 1 significant figure

12. A car travels 412 km on 38.7 litres of petrol. Estimate how far it travels on each litre.

13. Lani rounds 7.97 to 1 decimal place and writes 8. What is wrong with her answer?

★ Extension

14. A length is 12 cm, rounded to the nearest centimetre. Write the smallest and largest values the length could be.

15. A mass is 7.3 kg, rounded to 1 decimal place. Write the smallest and largest values the mass could be.

16. A rectangle measures 8 cm by 5 cm, each rounded to the nearest centimetre. Find the smallest area it could have.

17. A number is 5,200 when it is rounded to 2 significant figures. What is the smallest the number could be?

18. How many whole numbers give 900 when they are rounded to the nearest hundred?

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